ANIMAL SPECIES IDENTIFICATION SYSTEM USING IMAGE RECOGNITION AND DEEP LEARNING MODELS

Problem Statement:

Identifying animal species accurately is important for wildlife conservation, biodiversity research, veterinary services, and ecological monitoring. Traditional identification methods require expert knowledge and manual observation, making the process time-consuming and prone to human error. Variations in animal appearance, lighting conditions, and image quality further increase the difficulty of accurate species identification.

The proposed solution is an **Animal Species Identification System** that automatically recognizes animal species from images using deep learning and computer vision techniques. The system provides a centralized platform where users can upload animal images and receive prediction results through a secure and user-friendly interface.

By integrating Convolutional Neural Networks (CNNs) and image classification algorithms, the system extracts visual features from uploaded images and accurately classifies different animal species. It supports image preprocessing, feature extraction, real-time prediction, and performance evaluation while generating classification reports and confidence scores. Administrators can monitor model accuracy and manage image datasets efficiently.

The system is secure, scalable, and suitable for wildlife researchers, zoological organizations, veterinary institutions, and educational organizations.

Ultimately, this project delivers a modern and intelligent animal identification solution that improves species classification accuracy, supports wildlife conservation, reduces manual effort, and enhances biodiversity monitoring.